

Throughout this section, κ denotes the membrane permeability, T the target-hit time, τ_{out} the first exit time from the corridor band, and τ_{mem} the first genuine membrane crossing. The three windows **peak1**, **valley**, and **peak2** are compared using two complementary descriptions: *normalized* observables, reported as ratios or percentages of T , and *absolute* budgets, reported in expected step counts. More specifically, for an event time τ , the relative timing ratio is

$$\frac{E[\tau \mid \tau < T, T \in W]}{E[T \mid T \in W]},$$

i.e., the mean event time expressed as a fraction of the mean hit time in window W ; the outside-time share is the fraction of T spent outside the corridor band among trajectories whose hit times fall in the corresponding window; the outside budget is the expected number of pre-hit steps spent outside the corridor band; and the post-crossing budget is the expected number of steps remaining after the first membrane crossing and before target hit. In Figure 1, “outside” is the union of the left-side pocket and the merged outer/right-side bulk.

Figure 1: Top: spatial partition for the symmetric one-target baseline. Bottom: hit-time curves $f(t)$ for $\kappa = 0$ and $\kappa = 0.0040$, with embedded stacked bars for the **peak1**, **valley**, and **peak2** windows marked on the curves. The three segments give the normalized time shares in the left side, corridor, and merged outer/right side. The t_1 , t_v , and t_2 markers indicate the true window centres; the bars are shifted horizontally only for legibility.

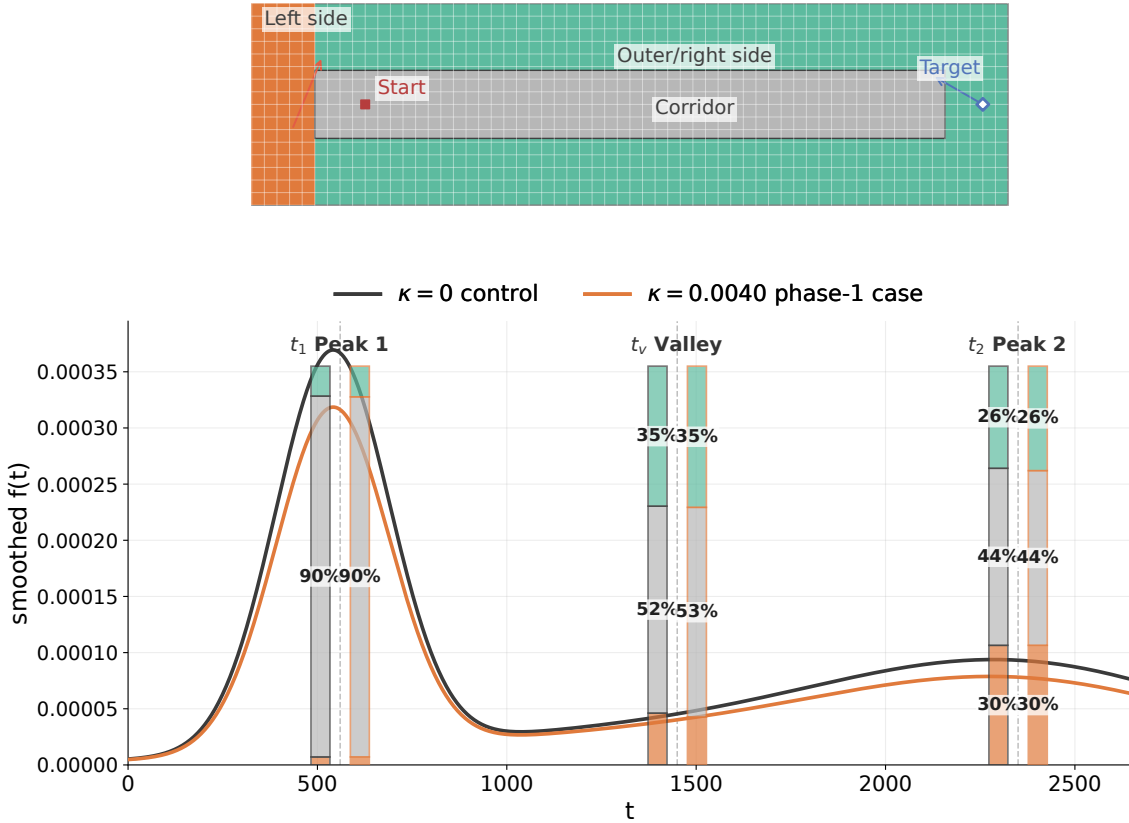


Figure 1 first locates the late-time structure in space. The curve separates the direct **peak1** arrivals from a valley and a later **peak2** window, but this visual feature is not labelled **double_peak** unless the quantitative classifier supports that label; weaker cases should remain shoulders or local bumps. The budget bars show why the later window is delayed. The outside share rises from about 48% to 56% at $\kappa = 0$ and from about 47% to 56% at $\kappa = 0.0040$, with the left-side pocket reaching about 30% in the later window. Thus the second-peak window is not just a plotting artefact: it is enriched in paths that spend extra pre-hit time outside the corridor, especially in the left-side pocket, before returning to the target. This additional residence time supplies the extra time scale behind the late-time feature.

Figure 2: Relative timing ratio for τ_{out} , defined as $E[\tau_{\text{out}} \mid \tau_{\text{out}} < T, T \in W]/E[T \mid T \in W]$, that is, the mean exit time expressed as a fraction of the mean hit time in window W . Black labels give the relative timing ratios, colour-matched percentages give the outside-time shares, and the top-right annotation gives the corresponding outside budgets, namely the expected numbers of pre-hit steps spent outside the corridor band, for the valley and **peak2** windows. The three series correspond to $\kappa = 0$, $\kappa = 0.0040$, and the boundary value $\kappa = 0.0152$.

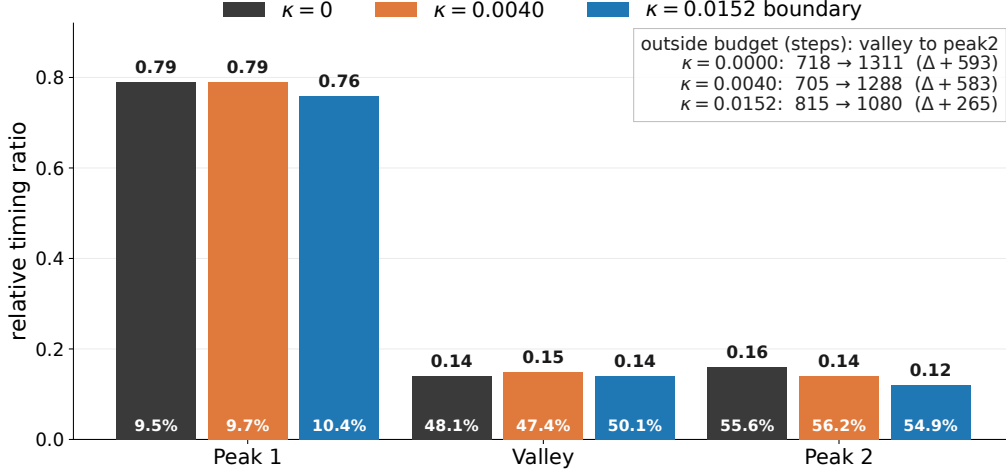


Figure 2 shows that the timing of the first corridor exit is only part of the story. **peak1** behaves as the direct-corridor benchmark: its exit ratio is about 0.76–0.79 and its outside-time share is only 9%–10%. The valley and **peak2** windows, however, both have small exit ratios, so the normalized exit time alone does not explain why one window lies later. The separation appears in the absolute outside budget: it increases from 718 to 1311 steps at $\kappa = 0$ and from 705 to 1288 steps at $\kappa = 0.0040$. The mechanism is therefore not simply “early exit”, but long residence after exit. Together with Figure 1, this points to outside excursions, especially through the left-side pocket, as the source of the delayed mass in the full first-passage distribution rather than a statement about the mean first-passage time alone.

Figure 3: Relative timing ratio for τ_{mem} , i.e., the mean first-crossing time expressed as a fraction of the mean hit time in window W . Black labels give the relative timing ratios, colour-matched percentages give the membrane-crossing probabilities $P(\tau_{\text{mem}} < T \mid T \in W)$, and the top-right annotation gives the post-crossing budgets, namely the expected numbers of steps remaining after the first crossing and before target hit. The label ‘n/a’ denotes that the ratio is undefined when no crossing occurs before target hit. The three series correspond to $\kappa = 0$, $\kappa = 0.0040$, and the boundary value $\kappa = 0.0152$.

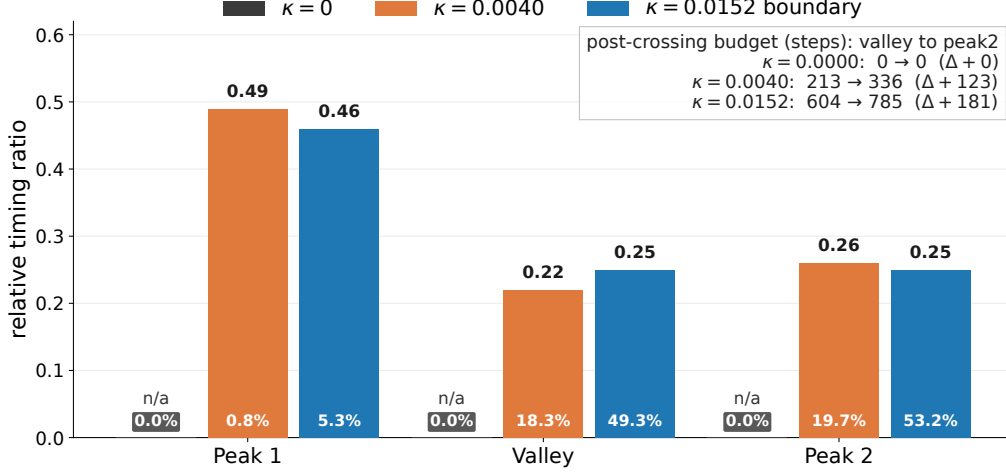


Figure 3 separates the membrane-mediated part of the same delay mechanism. When $\kappa = 0$, crossing and post-crossing budgets vanish, as they should. Once $\kappa > 0$, the valley and **peak2** windows are still close in crossing probability, about 18% versus 20% at $\kappa = 0.0040$, but they differ more clearly in the time left after the first crossing: about 213 versus 336 steps, increasing to about 604 and 785 steps at $\kappa = 0.0152$. Permeability therefore does not act mainly by making crossing more frequent. Its visible effect is to enlarge the post-crossing budget, so the trajectories that enter this channel still have a longer remaining path to absorption. This supports a delayed-residence explanation for the shoulder or second-peak contribution, while any **double_peak** label remains a classifier-level claim.